Final Metrics Report

Privacy-Preserving Medical Diagnosis using Federated Learning (pneumonia detection from chest X-rays)

1. System and evaluation protocol

The deployed model is DenseNet121 (ImageNet weights, backbone frozen) with a trainable head of about 263 thousand parameters, trained by FedAvg across three simulated hospitals (Kermany as Hospital A, two RSNA shards as B and C) with differential privacy (DP-SGD, $\varepsilon = 4$, $\delta = 10^{-5}$, RDP accountant), Secure Aggregation (SecAgg+) and TLS with client authentication. Splits are made at patient level. All figures below are measured on the held-out pooled test set, with model selection, temperature (2.2534) and decision threshold (0.35) fixed on validation data only.

Evaluation cohort. The primary metrics score the pneumonia-versus-normal task on 3,077 test images. They exclude 1,761 RSNA images that the dataset labels Normal but which show other abnormalities (the "No Lung Opacity / Not Normal" group). Results on all 4,838 test images are given beside them in every table and are the figures that describe use on unselected patients.

2. Headline performance (deployed checkpoint, threshold 0.35)

Table 1. Classification metrics, seed-42 deployed checkpoint

Metric	Primary: clear-label cohort (n = 3,077)	All test cases (n = 4,838)
AUROC (95% bootstrap CI)	**0.884** (0.872–0.896)	0.787 (0.774–0.800)
AUPRC	0.892	0.599
Accuracy	**81.2%**	72.5%
Sensitivity (recall)	73.1%	73.1%
Specificity	89.0%	72.2%
Precision (PPV)	86.3%	54.2%
Negative predictive value	77.6%	85.6%
F1 score	0.792	0.623
Balanced accuracy	81.0%	72.7%
TP / FP / FN / TN	1098 / 174 / 404 / 1401	1098 / 927 / 404 / 2409

Sensitivity is identical in both columns because no pneumonia case is removed. The difference comes entirely from false positives: the excluded group is flagged as pneumonia 42.8% of the time (RSNA images that are truly normal: 9.3%), accounting for 753 of the 927 false positives on the full set. The same model scored over three seeds on all cases gives AUROC 0.800 ± 0.012, sensitivity 73.9% ± 1.1, specificity 73.2% ± 1.9, F1 0.633 ± 0.011, ECE 0.026 and Brier score 0.164.

3. Human-in-the-loop layers

Predictions within ±0.05 of the threshold are reported as inconclusive, and the 10% of cases with the highest Monte Carlo Dropout entropy (T = 20 passes, cutoff 0.673) are deferred to a clinician.

Table 2. Effect of abstention and deferral (clear-label cohort / all cases)

Stage	Cases kept	Accuracy	Sensitivity	Specificity
No review layer	100% / 100%	81.2% / 72.5%	73.1% / 73.1%	89.0% / 72.2%
After inconclusive band	87.8% / 87.3%	84.6% / 75.2%	76.1% / 76.1%	92.5% / 74.8%
After band and deferral	78.8% / 77.1%	84.5% / 78.7%	70.7% / 70.7%	95.2% / 81.8%

4. Cost of each privacy layer (macro AUROC, 3 seeds, test split)

Table 3. Ablation ladder, macro (within-hospital) test AUROC, mean ± sample std

Configuration	Macro AUROC	Sens. at 90% spec.	Reading
Centralized pooled	0.893 ± 0.001	0.686 ± 0.013	Privacy-free ceiling
FedAvg (project recipe)	0.876 ± 0.006	0.638 ± 0.020	Federation costs about 0.017
FedAvg, DP-matched recipe	0.869 ± 0.005	0.603 ± 0.012	Baseline for the DP rows
FedAvg + SecAgg+ (real pipeline)	0.865 ± 0.012	0.616 ± 0.040	No measurable loss
FedAvg + DP, ε = 8	0.839 ± 0.008	0.556 ± 0.023	
FedAvg + DP, ε = 4 (target of the app)	0.830 ± 0.010	0.538 ± 0.018	Costs about 0.038 AUROC, 0.065 sensitivity
FedAvg + DP, ε = 2	0.817 ± 0.013	0.509 ± 0.014	
FedAvg + DP, ε = 1	0.796 ± 0.018	0.476 ± 0.030	Strongest privacy
Row 6: FedAvg + SecAgg+ + DP, ε = 4	0.830 ± 0.010	0.538 ± 0.018	3 seeds, simulation

Federation costs about 0.017 macro AUROC; secure aggregation costs nothing measurable; differential privacy adds a further, monotonic loss that falls hardest on the smallest hospital. The pooled AUROC used in an earlier analysis (0.905 centralized, 0.814 FedAvg) mixes hospitals with different prevalence and overstated the federation cost. The DP rows use the validation-selected recipe; the deployed app model is the original-recipe checkpoint.

Table 4. Calibration, privacy leakage and explanation quality (3 seeds)

Quantity	Result
ECE: no DP / with DP (ε = 1 to 8)	0.02–0.03 / 0.094–0.104 (about 4×); about 0.03 after temperature scaling
Conformal prediction (90% target, ε = 4)	Empirical coverage 0.908, mean set size 1.36
Membership inference (ε = 4)	Loss attack 0.4966 ± 0.0018; unprotected centralized 0.5084 against its negative control 0.5038, so no detectable leakage anywhere. Shadow-model (LiRA) attack: 0.517 without DP, 0.499 at ε = 4 (chance 0.5)
Grad-CAM pointing game / IoU (ε = 4)	0.147 ± 0.058 / 0.084 ± 0.038 (centralized 0.184 / 0.099)
Out-of-distribution flag rate (in-distribution test)	4.6–6.5% against a 5% target
Dirichlet non-IID FedAvg, α = 0.1 / 1.0	0.776 / 0.895 AUROC

5. Overhead

Table 5. Communication and compute (seed 42)

Quantity	Value
Payload per client per round (all layers)	1,053,853 bytes (about 1 MB; full DenseNet121 about 28 MB)
Last-round wall-clock, without / with DP	0.52 s / 6.74 s
Secure aggregation overhead	SecAgg+: 2.10 MB upload and 4.16 MB download per client per round (about 2 x plain), control messages about 2 KB; 83 s for 20 rounds on the GPU
Tests	293 automated tests, all passing

6. Limitations

The operating-point figures come from one seed's checkpoint. Hospitals are simulated on one machine, there is no external validation set, and the membership-inference attack used is weak, so it supports but does not prove the privacy claim. The primary cohort removes the cases on which the model is weakest, and the all-cases column should be read as the performance on unselected patients. A 3-seed experiment that retrained the model on the filtered data raised clear-label AUROC from 0.907 to 0.930 but lowered all-case AUROC from 0.809 to 0.780 (specificity 88% to 62%), so the deployed model was kept. This is a research prototype and is not validated for clinical use.